

Hariprakash Karthikeyan

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WORK EXPERIENCE

Data Scientist - Marketing Butler/Till

August 2025 - December 2025
Rochester, NY

- Replaced unstable DMA-level Media Mix Modeling (MMM) ($R^2 = 0.55$) with a clustered hierarchical model across 188 markets, improving validation R^2 of +13.7% and reducing allocation volatility driven by noisy local estimates.
- Stabilized regional media effect estimates (cluster R^2 0.88–0.92) using partial pooling, enabling consistent comparison of channel performance across markets and reducing contradictory budget recommendations.
- Automated nonlinear adstock + Hill-based modeling workflows (PyMC + JAX), reducing scenario analysis time by 40–50% and increasing the number of budget simulations evaluated per planning cycle.
- Identified channel-specific carryover and saturation behavior (e.g., TV high carryover; Display early diminishing returns), supporting more disciplined incremental spend decisions and avoidance of oversaturated allocations.
- Validated model stability across planning cycles by stress-testing hierarchical priors and adstock parameterizations, ensuring consistent budget recommendations every quarter and eliminating the need for costly mid-cycle model re-runs by planning teams without retraining instability.

Data Science Intern Buffalo Solar

June 2025 - July 2025
Buffalo, NY

- Engineered a Python-based monitoring pipeline using Pandas and REST API (SolarEdge) to process 15-minute production telemetry across 468 solar sites, generating prioritized anomaly alerts and eliminating manual performance review.
- Productionized time-series forecasting and anomaly detection workflows into an interactive Streamlit + Plotly dashboard adopted by engineering and operations teams, reducing manual monitoring effort by ~30–40%.
- Improved anomaly detection precision to ~90% through residual analysis, threshold tuning, and statistical filtering, lowering false inspection flags and improving maintenance prioritization.
- Designed modular ETL and aggregation logic for fleet-level analytics (monthly/yearly energy trends, outage filtering, MAE/RMSE tracking), enabling faster identification of underperforming systems.

Research Data Scientist University of Rochester Medical Center

February 2025 - May 2025
Rochester, NY

- Built longitudinal EHR processing pipelines in Python (Pandas, NumPy) to filter and merge 2,000+ patient records across admissions, replacing manual chart review with reproducible cohort construction.
- Implemented regex-based NLP using Python's re module to extract structured phenotypes (e.g., LVEF values) from unstructured clinical notes, transforming fragmented documentation into an 845-patient analysis-ready dataset.
- Constructed time-aligned feature tables by integrating labs, admissions, and note-derived variables, enabling reusable datasets for downstream statistical modeling and research analyses.

DATA SCIENCE PROJECTS

[Real-Time Music Data Platform & LLM Assistant | AWS S3, Kafka, LangChain, OpenAI, Python, Docker](#)

- Orchestrated a cloud-native music data platform (S3, Kafka, SQL, Python, Docker) processing 500K+ events with idempotent deduplication, enabling scalable user-behavior analytics and retention-focused recommendation workflows.
- Integrated a low-latency (less than 2s) SQL-grounded LLM music assistant (LangChain, pgvector) to deliver personalized recommendations designed to improve session continuity and reduce skip rates.

[End-to-End Distributed Sentiment Analytics | AWS \(Databricks\), Delta Lake, MLflow](#)

- Architected a Spark Structured Streaming medallion pipeline on AWS-hosted Databricks to ingest 100K+ tweets, operationalizing transformer-based sentiment inference via MLflow-packaged models and Spark UDFs backed by Delta Lake for fault-tolerant, reproducible near real-time analytics.

[Solder Joint Defect Detection | Deep Learning, TensorFlow, Transfer Learning](#)

- Fine-tuned MobileNetV2 and EfficientNetV2 on solder images with panel-level data splits to prevent leakage, achieving ~0.80 accuracy and ~0.95 defect recall via A/B threshold tuning, reducing missed defects in automated quality inspection.

SKILLS

- Programming & Query Languages:** Python, SQL
- Libraries & Frameworks:** Pandas, NumPy, scikit-learn, TensorFlow, PyMC, JAX, LangChain
- Visualization & Apps:** Plotly, Matplotlib, Seaborn, Streamlit, FastAPI
- Statistical Methods:** Bayesian Modeling, Time-Series Forecasting, A/B Testing, Causal Inference, Predictive Modeling
- Cloud & Data Platforms:** AWS (S3, EC2, ECR), PostgreSQL, Git, Linux
- Big Data & Streaming:** Spark (PySpark, Structured Streaming), Kafka, Databricks, Delta Lake, Dask
- NLP & LLM:** Transformers, Embeddings, RAG, pgvector, Regex NLP
- MLOps & Versioning:** MLflow, DVC, Docker, GitHub Actions

EDUCATION

University of Rochester
Master of Science in Data Science

Rochester, NY
August 2024 - December 2025

SRM Institute of Science and Technology (SRMIST)
Bachelor of Technology in Electronics and Communication Engineering

Chennai, India
August 2020 - May 2024